

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 
□□□□□□□□

02 / 
1NF    → 2NF → 3NF

03 / 
JOIN □□□

04 / 
OLTP vs OLAP □□□

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PART 01



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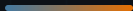
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データが正常に更新された

B データが正常に更新されなかった

C データが正常に更新された

PART 02



1NF → 2NF → 3NF → ☐ ☐ ☐ ☐ ☐ ☐

1NF

NO DUPLICATES

BEFORE — 1NF

student	courses
101	Math, Science, History, English, Art, Music, PE

AFTER — 1NF

student	course
101	Math
101	Science
101	History
101	English
101	Art
101	Music
101	PE

1NF

2NF

3NF

2NF

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO SLIP

□□□□□□

student_id → dept_id → dept_name — dept_name — dept_id —

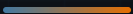
BEFORE — □□□□

student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□



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PART 03



□□□□□□JOIN □□□

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JOIN □□
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JOIN □□□□□

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OLTP vs OLAP

OLTP	OLAP	Comparison
Transactional	Analytical	
High write volume	High read volume	
Simple queries	Complex queries	
Real-time processing	Batch processing	
Normalized database	Denormalized database	
OLTP	OLAP	

STRATEGY

OLTP is designed for high transaction volume and fast response time. OLAP is designed for high data volume and complex queries.



→



1NF                                               

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OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX